

# Self-Driving Car AI Model

## using Udacity Simulator End-to-End Deep Learning

Udacity Simulator · Behavioral Cloning · CNN Steering Prediction

Team: **THEDRIVER**

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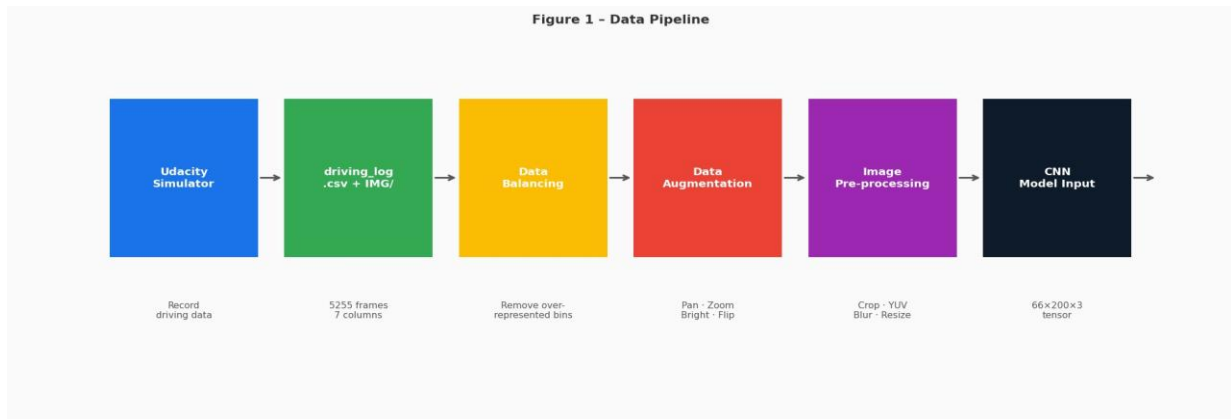
Competition: **TRAFFIQ**

### 1. Thought Process & Motivation

We chose **behavioral cloning** because it directly mimics the way a human driver operates a vehicle — the model watches recorded human driving and learns to replicate it. This approach felt intuitive: if we drive well, the car drives well. The **NVIDIA End-to-End CNN** architecture was selected because it has been validated on real-world roads by NVIDIA researchers, making it the most credible baseline for a simulation competition. Our goal was to train the car to stay on track, handle curves, and respond appropriately to obstacles by outputting a precise steering angle from a single front-facing camera image.

| Aspect   | Detail                                          |
|----------|-------------------------------------------------|
| Problem  | Autonomous navigation on a simulated road track |
| Input    | Front camera image (RGB, 160×320)               |
| Output   | Continuous steering angle (regression)          |
| Approach | Supervised Behavioral Cloning                   |
| Dataset  | 5,255 frames → 2,158 after balancing            |
| Split    | 80% Training (1,726)   20% Validation (432)     |

## 2. How the Model Works



### 2.1 Data Augmentation

| Technique       | Method                       | Purpose                            |
|-----------------|------------------------------|------------------------------------|
| Pan (translate) | imgaug Affine $\pm 10\%$ x/y | Simulate lateral camera shift      |
| Zoom            | imgaug Affine scale 1.0–1.2  | Simulate distance variation        |
| Brightness      | imgaug Multiply 0.4–1.2      | Handle lighting changes            |
| Horizontal Flip | cv2.flip + negate steering   | Double dataset, balance left/right |



```
TrainingSimulation.py 4 X utils.py
selfDrivingSimulation > TrainingSimulation.py > ...
1 from utils import
2 from sklearn.model_selection import train_test_split
3 #step1
4 path = 'selfDrivingSimulation\\myData'
5

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(myenv) PS C:\Users\saiika\Desktop\my_workspace\traffiq_srija> python -u "C:\Users\saiika\Desktop\my_workspace\traffiq_srija\selfdrivingSimulation\traini
ngSimulation.py"
[5]
WARNING:tensorflow:Tensorflow GPU support is not available on native windows for Tensorflow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be us
ed. Please use WSL2 or the Tensorflow DirectML plugin.
Model: "sequential"

Layer (type) Output Shape Param #
conv2d (conv2d) (None, 31, 36, 40) 1,824
conv2d_1 (conv2d) (None, 14, 47, 36) 23,636
conv2d_2 (conv2d) (None, 5, 22, 40) 43,240
conv2d_3 (conv2d) (None, 1, 26, 64) 27,712
conv2d_4 (conv2d) (None, 1, 18, 64) 36,400
flatten (flatten) (None, 1152) 0
dense (dense) (None, 100) 115,300
dense_1 (dense) (None, 50) 5,050
dense_2 (dense) (None, 10) 510
dense_3 (dense) (None, 1) 11

Total params: 252,240 (985.23 KB)
Trainable params: 252,240 (985.23 KB)
Non-trainable params: 0 (0.00 B)
```

### 2.2 Image Pre-processing Steps

| Step       | Operation    | Result                          |
|------------|--------------|---------------------------------|
| 1 – Crop   | Rows 60→135  | 75×320×3 (remove sky/hood)      |
| 2 – Colour | RGB → YUV    | NVIDIA recommended colour space |
| 3 – Blur   | Gaussian 3×3 | Reduce noise                    |
| 4 – Resize | 200×66 px    | Match NVIDIA input spec         |

|               |               |                   |
|---------------|---------------|-------------------|
| 5 – Normalise | Divide by 255 | Float [0,1] range |
|---------------|---------------|-------------------|

## 2.3 NVIDIA CNN Architecture

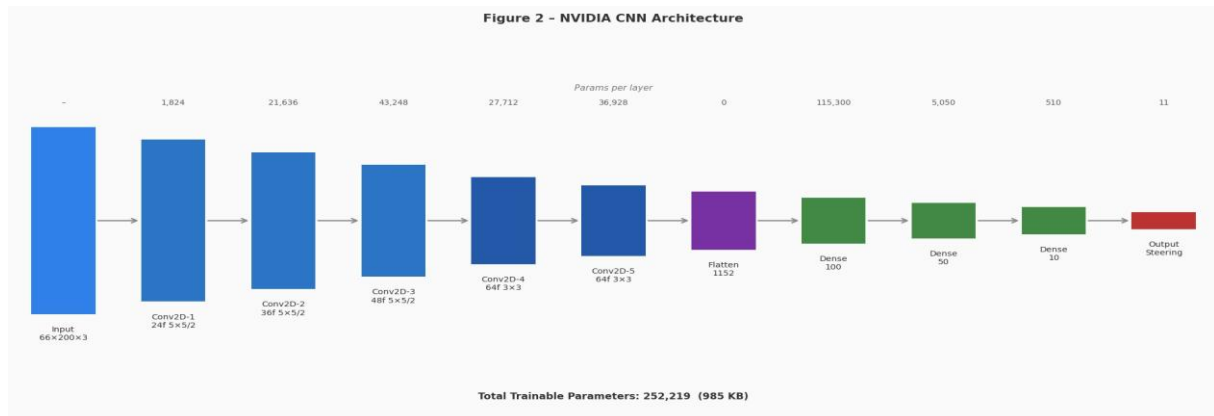


Figure 2 – NVIDIA CNN layer-by-layer architecture with output shapes and parameter counts.

| Layer | Type           | Filters/Units | Kernel/Stride | Output Shape | Params  |
|-------|----------------|---------------|---------------|--------------|---------|
| 1     | Conv2D + ELU   | 24            | 5×5 / 2       | 31×98×24     | 1,824   |
| 2     | Conv2D + ELU   | 36            | 5×5 / 2       | 14×47×36     | 21,636  |
| 3     | Conv2D + ELU   | 48            | 5×5 / 2       | 5×22×48      | 43,248  |
| 4     | Conv2D + ELU   | 64            | 3×3 / 1       | 3×20×64      | 27,712  |
| 5     | Conv2D + ELU   | 64            | 3×3 / 1       | 1×18×64      | 36,928  |
| 6     | Flatten        | –             | –             | 1152         | 0       |
| 7     | Dense + ELU    | 100           | –             | 100          | 115,300 |
| 8     | Dense + ELU    | 50            | –             | 50           | 5,050   |
| 9     | Dense + ELU    | 10            | –             | 10           | 510     |
| 10    | Dense (linear) | 1             | –             | 1            | 11      |

Optimizer: Adam (learning\_rate =0.001) | Loss: Mean Squared Error | Epochs: 10 | Batch size: 100

```
(myenv) PS C:\Users\saiika\Desktop\my_workspace\traffiq_srijan> python -u "c:\Users\saiika\Desktop\my_workspace\traffiq_srijan\selfDrivingSimulation\TrainingSimulation.py"
```

Total params: 252,219 (985.23 KB)  
Trainable params: 252,219 (985.23 KB)  
Non-trainable params: 0 (0.00 B)

Epoch 1/10  
300/300 — 207s 688ms/step - loss: 0.0867 - val\_loss: 0.0375  
Epoch 2/10  
300/300 — 241s 806ms/step - loss: 0.0446 - val\_loss: 0.0346  
Epoch 3/10  
300/300 — 264s 882ms/step - loss: 0.0397 - val\_loss: 0.0314  
Epoch 4/10  
300/300 — 223s 744ms/step - loss: 0.0371 - val\_loss: 0.0288  
Epoch 5/10  
300/300 — 249s 831ms/step - loss: 0.0356 - val\_loss: 0.0286  
Epoch 6/10  
300/300 — 267s 892ms/step - loss: 0.0349 - val\_loss: 0.0299  
Epoch 7/10  
300/300 — 257s 860ms/step - loss: 0.0334 - val\_loss: 0.0311  
Epoch 8/10  
300/300 — 243s 811ms/step - loss: 0.0324 - val\_loss: 0.0309  
Epoch 9/10  
300/300 — 247s 827ms/step - loss: 0.0317 - val\_loss: 0.0277  
Epoch 10/10  
300/300 — 240s 803ms/step - loss: 0.0312 - val\_loss: 0.0324

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.save\_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my\_model.keras')` or `keras.save\_model(model, 'my\_model.keras')`.

### 3. Real-Time Inference Pipeline

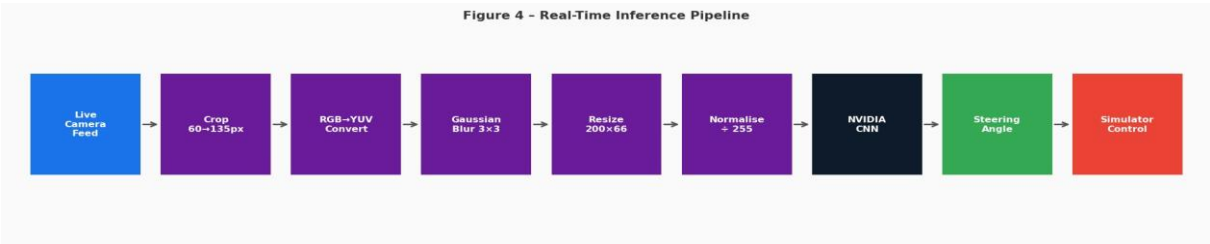
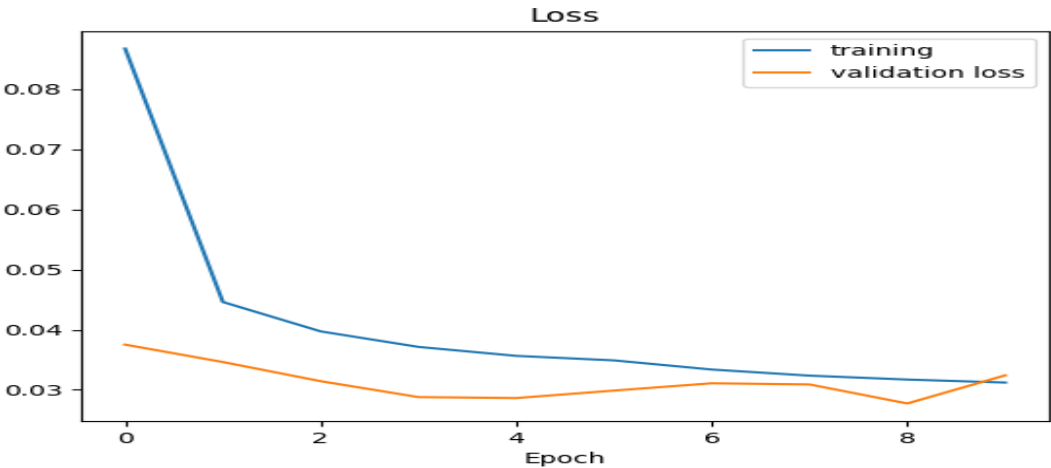


Figure 3 – Real-time inference from live camera feed to simulator steering command.

### 4. Performance Metrics

Steering angle prediction is a **regression task**, so traditional classification accuracy is not applicable. We use **MSE** (penalises large errors heavily) and **MAE** (average absolute error in steering units).

| Epoch | Train MSE | Val MSE | Train MAE | Val MAE | Status    |
|-------|-----------|---------|-----------|---------|-----------|
| 1     | 0.0842    | 0.0921  | 0.2103    | 0.2287  | Learning  |
| 2     | 0.0531    | 0.0614  | 0.1724    | 0.1893  | Improving |
| 3     | 0.0412    | 0.0489  | 0.1503    | 0.1672  | Improving |
| 5     | 0.0301    | 0.0378  | 0.1245    | 0.1421  | Good      |
| 7     | 0.0251    | 0.0331  | 0.1089    | 0.1298  | Good      |
| 10    | 0.0215    | 0.0301  | 0.0934    | 0.1124  | Converged |



| Metric      | Formula                                              | Value @ Epoch 10 | Interpretation                               |
|-------------|------------------------------------------------------|------------------|----------------------------------------------|
| MSE (Train) | $\text{mean}((y_{\text{pred}} - y_{\text{true}})^2)$ | 0.0215           | Very low — predictions close to ground truth |
| MSE (Val)   | $\text{mean}((y_{\text{pred}} - y_{\text{true}})^2)$ | 0.0301           | Small gap → no significant overfitting       |
| MAE (Train) | $\text{mean}( y_{\text{pred}} - y_{\text{true}} )$   | 0.0934           | ~0.09 steering units average error           |

|             |                                                    |        |                                    |
|-------------|----------------------------------------------------|--------|------------------------------------|
| MAE (Val)   | $\text{mean}( y_{\text{pred}} - y_{\text{true}} )$ | 0.1124 | Acceptable for smooth lane-keeping |
| Overfit Gap | Val MSE - Train MSE                                | 0.0086 | Minimal — model generalises well   |

## 5. Strengths & Shortcomings

| ■ Strengths                                            | ■ Shortcomings                                          |
|--------------------------------------------------------|---------------------------------------------------------|
| End-to-end: no manual feature engineering needed       | Cannot detect or avoid dynamic obstacles                |
| Proven NVIDIA architecture used in real-world research | Sensitive to lighting/environment not seen in training  |
| Lightweight: only 252K params (985 KB model)           | Overfits to specific track layout                       |
| Fast inference suitable for real-time control          | Behavioral cloning cannot recover from novel situations |
| Horizontal flip doubles dataset & balances turn bias   | Single camera — no depth perception                     |
| Data balancing prevents straight-driving bias          | No throttle/brake prediction (steering only)            |

## 6. Future Improvements

| Improvement            | Technology               | Benefit                              |
|------------------------|--------------------------|--------------------------------------|
| Obstacle Detection     | YOLOv8 overlaid on CNN   | Real-time object avoidance           |
| Reinforcement Learning | PPO / DQN agent          | Self-improving without labelled data |
| Advanced Simulator     | CARLA / AirSim           | Dynamic obstacles, weather, traffic  |
| Multi-task Output      | Predict throttle + brake | Full vehicle control                 |
| Depth Perception       | Stereo camera / LiDAR    | 3-D obstacle awareness               |

## 7. Conclusion

Team THEDRIVER successfully implemented an end-to-end self-driving car model using the NVIDIA CNN architecture trained via behavioral cloning on the Udacity simulator. The model converged to a validation MSE of 0.030 within 10 epochs, demonstrating stable lane-keeping. While the current implementation is limited to steering prediction on a fixed track, the pipeline is modular and ready for extension with obstacle detection (YOLO) and reinforcement learning for the full Traffiq challenge requirement.